An Automated Pipeline for Short-Form Video: Speech Recognition, Word-Synchronous Captions, Conditional Music Generation and Distribution

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Abstract

We describe the mathematics behind a pipeline that turns a raw video into a captioned, scored video published on YouTube, TikTok and Instagram. Speech is transcribed by Whisper, an encoder–decoder Transformer on log-Mel spectrograms; word timestamps are obtained by dynamic time warping on cross-attention; captions are formed by an optimal segmentation of the word sequence and highlighted word by word with drift-free rounding. Background music is sampled from MusicGen, an autoregressive Transformer over residual-vector-quantised audio codes with classifier-free guidance, conditioned on a prompt extracted from the transcript, and mixed at a target loudness. Distribution is a scheduling problem under API quotas.

1. Speech to text

Features. The audio $x[n]$ is resampled to 16 kHz; its short-time Fourier transform with a 25 ms window w and a 10 ms hop h, $X(t,f)=\sum_n x[n]\,w[n-th]\,e^{-2\pi i f n/N}$, is projected on 80 triangular filters Φ_k equally spaced on the Mel scale $\mathrm{mel}(f)=2595\log_{10}(1+f/700)$ [11], giving $S(t,k)=\log\big(\sum_f \Phi_k(f)|X(t,f)|^2+\epsilon\big)$.

Model. Whisper [7] is a Transformer encoder–decoder [12]: the encoder maps 30 s windows of S to states E, the decoder defines $p_\theta(y_u \mid y_{<u}, E)$ over text tokens, including special tokens for language, task (transcribe/translate) and time. It was trained by cross-entropy on 680,000 hours of weakly supervised audio, which explains its robustness across accents and languages. Decoding approximates $\arg\max_y p_\theta(y \mid S)$ by beam search, with temperature fallback when the log-probability or the compression ratio of the output signals a failure.

Word timestamps. Let $A \in \mathbb{R}^{U\times F}$ be the cross-attention of selected heads between the U output tokens and the F audio frames (normalised per row). Since speech and text are both ordered in time, we seek a monotone path $\gamma=((u_\ell,f_\ell))_\ell$ from $(1,1)$ to (U,F) with steps in $\{(1,0),(0,1),(1,1)\}$ minimising $\sum_\ell -A_{u_\ell f_\ell}$. Dynamic time warping [10] solves it exactly in $O(UF)$:

$$D(u,f) = -A_{uf} + \min\{D(u-1,f),\, D(u,f-1),\, D(u-1,f-1)\}. \quad (1)$$

The first and last frames matched to the tokens of a word give its interval $[s_i, e_i]$.

2. Captions

Segmentation. Given words $w_1,\dots,w_n$ with intervals $[s_i,e_i]$, we formalise the grouping of words into captions: a caption is a block $w_{a..b}$ of consecutive words; it is admissible if its length is at most L characters, its duration e_b-s_a at most $\tau_{\max}$ and no pause inside exceeds δ. For the cost $c(a,b)=1+\kappa\,(L-\mathrm{len}(w_{a..b}))^2$, which favours few, well-filled captions, the optimal segmentation is computed by

$$V(b)=\min_{a\le b,\ w_{a..b}\ \text{admissible}} V(a-1)+c(a,b),\quad V(0)=0, \quad (2)$$

in $O(nK)$ where K is the maximal number of words per caption; Bellman's principle gives optimality, as for line breaking. Captions are centred on screen, where the eye rests in vertical video.

Word highlighting. In the ASS format, a karaoke tag $\backslash \mathtt{k} d_i$ highlights word i for d_i centiseconds. Rounding each duration independently, $d_i=\mathrm{round}(100(e_i-s_i))$, accumulates an error up to $n/2$ cs over n words. Cumulative rounding, $d_i=\mathrm{round}(100e_i)-\mathrm{round}(100s_1)-\sum_{j<i}d_j$ (gaps included), keeps the end of every word within 0.5 cs of its true time, whatever n.

3. Music generation

Audio tokens. EnCodec [2] encodes a 32 kHz waveform into latent vectors z_t at 50 Hz, quantised by residual vector quantisation [13]: $r_0=z_t$, $q_{t,k}=\arg\min_{c\in\mathcal{C}_k}\|r_{k-1}-c\|$, $r_k=r_{k-1}-q_{t,k}$, $k=1,\dots,K$, with $K=4$ codebooks of size 2048; each stage quantises the residual of the previous ones, so the error decreases with k.

Language model. MusicGen [1] models the K parallel streams with a single Transformer using a *delay* pattern: stream k is shifted by $k-1$ steps, so that at step t the model predicts $(q_{t,1}, q_{t-1,2},\dots,q_{t-K+1,K})$, each conditionally on all previous steps and on the text through cross-attention to a T5 encoding [8] of the prompt. The factorisation $\prod_t\prod_k p(q_{t-k+1,k}\mid \text{past}, \text{text})$ treats the K codes of a step as conditionally independent given the past, the price paid for one forward pass per 20 ms of audio.

Sampling. With conditional and unconditional logits ℓ_c,ℓ_u, classifier-free guidance [4] samples from

$$\ell=\ell_u+\gamma(\ell_c-\ell_u),\quad p(v)\propto\exp(\ell_v/T), \quad (3)$$

i.e. $p\propto p_u^{1-\gamma}p_c^{\gamma}$ at $T=1$, which sharpens adherence to the prompt for $\gamma>1$. Top-k keeps the k most likely codes; nucleus sampling [5] keeps the smallest set V_p with $\sum_{v\in V_p}p(v)\ge p$. Both trade diversity against artefacts.

Prompt and adaptation. The prompt is built from the first sentence of the transcript, so the music follows the theme of the video. A lighter model can be fine-tuned by cross-entropy on the token sequences of MAESTRO [3] (about 200 hours of piano), giving royalty-free music specialised to a style at lower cost.

Mixing. Speech s and music m are mixed as $y=s+g\,m$. Integrated loudness [6] is $L=-0.691+10\log_{10}$ of the gated mean square of the K-weighted signal, so scaling by g shifts it by $20\log_{10}g$ dB. Choosing $g=10^{(L_s-\Delta-L_m)/20}$ places the music Δ dB (typically 15–20) below the speech.

4. Distribution

Each platform exposes an upload API authorised by OAuth 2.0 [9]. A batch of V videos is assigned to slots t on platforms j by $x_{vjt} \in \{0, 1\}$, maximising $\sum x_{vjt}\, r_{jt}$, where r_{jt} is the expected early engagement of slot t on platform j, subject to one post per video and platform, $\sum_v x_{vjt} \le 1$ per slot, and the quota $\sum_{v,t\in\text{day}} c_j x_{vjt} \le Q_j$ (c_j units per upload out of a daily budget Q_j). This is an assignment problem with knapsack side constraints, solved greedily or by integer programming; the engagement rates r_{jt} are estimated from past posts.

5. Status

Implemented: Whisper transcription with word timestamps, centred word-highlighted ASS captions, MusicGen generation and FFmpeg mixing, uploads through the three APIs. Next: the optimal caption segmentation and drift-free rounding above, loudness-targeted mixing, the posting schedule and the fine-tuned music model.

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